

Central Poupard Numbers

Two OEIS conjectural statements, complete asymptotic sectors,
and a universal transition for shifted moments

Research article prepared for Vladimir Reshetnikov

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Abstract

Let $T_n = (2n+1)![z^{2n+1}] \tan z$ and let $A_n(c) = \sum_{k=0}^n \binom{n}{k} c^{n-k} T_k$. The specialization $A_n(1)$ is OEIS A125054, the central Poupard numbers. We prove the continued-fraction identity recorded there as a conjecture in December 2025, and prove the stronger congruence $A_n(1) \equiv 3 \pmod{9}$ for every $n \geq 1$, which settles the entry's valuation observation. Both proofs use elementary formal continued fractions.

The main analytic development is considerably broader. A positive integral representation yields an exact, convergent decomposition into odd-pole exponential sectors, an explicit remainder bound, and inverse-power expansions of every fixed sector to arbitrary order. We then prove a uniform, all-order endpoint–saddle expansion when the shift is $c = \tau n^2$. Its first-order transition occurs at $\tau_* = 0.262466543368419\dots$, but its finite-size coexistence window is displaced by a multiple of $(\log n)/n$. We obtain the limiting mixture law and the Gaussian fluctuations inside its macroscopic component. Positivity also yields a complex coexistence limit and a rigorously located array of polynomial zeros. A general theorem extends the transition to moments of y^p under any probability density with tail $Cy^\beta e^{-\alpha y}$, $p > 1$. Finally, we invert the fixed-shift growth using Lambert W , including an exponentially small displacement relative to an exact one-sector inverse.

All essential arguments are supplied. Classical generating functions, continued-fraction machinery, and the previously recorded leading asymptotic are explicitly distinguished from the extensions developed here. The accompanying code provides exact finite checks and high-precision numerical diagnostics; it is not a Lean formalization or an interval proof. Worldwide publication priority for the broader extensions has not been established.

[Write note, batch 7301, 1 October 2026.] This is the collection edition of the report, written into ProveIt's research-report collection from manuscript 47 of batch 7301 (provenance in Appendix C). Every label carries the prefix `pou::`; the `Makefile` now lives in `code/`, and `requirements.txt` and `BUILD_REPORT.json` in `data/`. Text added in the write is marked like this note; everything else is the delivered text. The OEIS statements quoted here (Bala's conjectured fraction, Chapoton's observation, Kotesovec's leading term) are as the authors inspected the entry on 1 October 2026; the intake did not re-fetch it.

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1 Scope, provenance, and the central results

The two concrete OEIS targets are the continued fraction labelled “Conjectured g.f.” by Peter Bala on 15 December 2025, and the observation by F. Chapoton on 2 August 2021 that the noninitial terms of A125054 appear to have 3-valuation one. Both labels were present in the entry inspected for this article [1]. The statements proved below are stronger than finite verification: the continued fraction is an identity of formal power series, and the congruence holds for every positive index. These are resolutions of the displayed statements, not a claim that no unpublished or unlocated proof previously existed.

The sequence is not introduced merely to manufacture an asymptotic example. It is the center of the Poupard triangle, part of the finite-difference calculus of tangent and secant enumeration [5, 2]. Its tangent-number representation makes it a particularly explicit test case for the competition between a large deterministic shift and the factorial growth of moments.

The present work continues the repository’s emphasis on exact identities, all-order expansions, inversion, and transparent verification boundaries. The inspected ProveIt snapshot is commit

ef66df0fdb8fc64cbc80756d622cab0e8e85e262.

In particular, the repository’s logarithmic-endpoint/Lambert-chart report [13] is a methodological precedent, not a theorem used as a hypothesis here. Its countable-action model is different from the shifted moment integrals below. No repository result, formalized or otherwise, is required for the proofs in this article. The repository was not modified.

[Write note, batch 7301, 1 October 2026.] The precedent cited as [13] is the transseries package `Analysis/Transseries/docs/series-and-transseries/Logarithmic_Critical_Endpoint_Lambert_Charts/`; nothing is imported from it. At the repository’s current state no other report or formal development treats A125054, A125053, A000182 or the shifted tangent moments $A_n(c)$. One constant recurs: the transition parameter $s_* = 2 + W_0(-2e^{-2})$ of (4) is the number called τ in the collection report on A277364 [15], where it solves $\tau/(1 - e^{-\tau}) = 2$ (the same equation $2 - s = 2e^{-s}$) and governs the omitted tail $B_n - a(n)$ of the truncated Stirling sum $a(n) = \sum_{k \leq n/2} S(n, k)$. The objects differ; see the note after the numerical values of s_* and τ_* in Section 7.

1.1 What is known, what is proved here

The tangent generating function, its classical Stieltjes fraction, the Poupard bivariate generating function, and the leading equivalent of A125054 are prior results [6, 5, 1]. The moment and Hankel consequences are included because they explain the structure; they are not advertised as discoveries. Our substantive extension consists of the exact-sector remainder control, the uniform two-contribution expansion, the moving coexistence window and its probability laws, the $p > 1$ universality theorem, the associated complex zero array, and the inversion formulas. These are proved in the article, but a targeted source search is not an exhaustive priority determination.

For orientation, the central statements are as follows. Put

$$\alpha = \frac{\pi}{2}, \quad T_n = (2n+1)! [z^{2n+1}] \tan z, \quad A_n(c) = \sum_{k=0}^n \binom{n}{k} c^{n-k} T_k, \quad a_n = A_n(1). \quad (1)$$

The structural proofs establish the conjectured fraction and

$$a_n \equiv 3 \pmod{9} \quad (n \geq 1). \quad (2)$$

For fixed $c \geq 0$ we obtain the exact sector expansion

$$A_n(c) = \frac{2\Gamma(2n+2)}{\alpha^{2n+2}} \sum_{\substack{q \geq 1 \\ q \text{ odd}}} q^{-2n-2} H_n(c\alpha^2 q^2), \quad H_n(z) = \sum_{j=0}^n \frac{(z/4)^j}{j!(n + \frac{1}{2})_j}. \quad (3)$$

For the large shift $c = \tau n^2$, the transition constant is characterized by

$$2 - s_* = 2e^{-s_*}, \quad 1 < s_* < 2, \quad \tau_* = \frac{s_*(2 - s_*)}{\alpha^2}. \quad (4)$$

In a precisely specified moving window, the normalized moment tends to $1 + e^{-w}$, rather than to a single saddle contribution. The coefficient procedures and uniform errors make these formulas usable beyond their leading terms.

1.2 Conventions about asymptotics and formal series

The symbol $\sim \sum_{r \geq 0} c_r n^{-r}$ means a Poincaré expansion: for every fixed M , truncation before n^{-M} has error $O(n^{-M})$. It does not assert convergence of the infinite inverse-power series. An exact positive sector decomposition and an inverse-power expansion of one sector are different assertions and will be kept separate.

All ordinary generating functions of the factorially growing sequences in this article are formal power series. Their radius of convergence is zero. An infinite continued fraction means its coefficient-wise stabilized formal expansion, not its evaluation at a small positive real number. All probability statements concern explicitly normalized positive weights.

2 From the Poupard triangle to tangent moments

The first rows of the relevant Poupard triangle are

$$\begin{array}{c} 1, \\ 1, 3, 1, \\ 5, 15, 21, 15, 5, \\ 61, 183, 285, 327, 285, 183, 61. \end{array}$$

To fix indexing without relying on these examples, write $h_{i,j}$ for its entry with $i + j = 2n$, $i, j \geq 0$, and set $h_{i,j} = 0$ when $i + j$ is odd. The classical bivariate exponential generating function is

$$Z(x, y) = \sum_{i,j \geq 0} h_{i,j} \frac{x^i y^j}{i! j!} = \cos(x - y) \sec^2(x + y). \quad (5)$$

This is the secant case of Foata and Han's generating-function theorem [5, Theorem 1.2].

For completeness, the algebraic content of this representation can also be checked directly. It is symmetric in x, y , its boundary value is $Z(x, 0) = \sec x$, and it satisfies

$$(\partial_x - \partial_y)^2 Z = -4Z. \quad (6)$$

Consequently its coefficients satisfy

$$h_{i+2,j} - 2h_{i+1,j+1} + h_{i,j+2} = -4h_{i,j}.$$

The two boundary values of each even row and its prescribed second differences determine that row uniquely: the difference of two solutions would be an affine function of the position, vanishing at both ends. Thus (5) characterizes the triangle with secant-number edges. The combinatorial interpretation is prior work; the following short operator calculation explains the central specialization used here.

Proposition 2.1. *The central entry is $h_{n,n} = A_n(1)$. More generally, the diagonal exponential coefficient of*

$$Z_c(x, y) = \cos(\sqrt{c}(x - y)) \sec^2(x + y)$$

is $A_n(c)$; the expression is a formal power series in c , so the square root creates no ambiguity.

Proof. Set $s = x + y$ and $d = x - y$. Then $\partial_x \partial_y = \partial_s^2 - \partial_d^2$. Since $\partial_d^2 \cos(\sqrt{c}d) = -c \cos(\sqrt{c}d)$,

$$(\partial_x \partial_y)^n Z_c(0, 0) = (\partial_s^2 + c)^n \sec^2 s \Big|_{s=0} = \sum_{k=0}^n \binom{n}{k} c^{n-k} T_k.$$

Here $(\sec^2 s)^{(2k)}(0) = (\tan s)^{(2k+1)}(0) = T_k$. The left side is exactly $n! [x^n y^n] Z_c(x, y)$. $\square$

One exact recurrence useful for checking the normalization comes from $\tan' = 1 + \tan^2$:

$$T_0 = 1, \quad T_n = \sum_{j=0}^{n-1} \binom{2n}{2j+1} T_j T_{n-1-j} \quad (n \geq 1). \quad (7)$$

It gives $T_n = 1, 2, 16, 272, 7936, \dots$ (OEIS A000182 [3]), hence $a_n = 1, 3, 21, 327, 9129, \dots$

3 A proof of the conjectured continued fraction

3.1 Formal contractions and binomial translation

For a sequence of weights $\gamma_1, \gamma_2, \dots$, write

$$S(\gamma; x) = \frac{1}{1 - \frac{\gamma_1 x}{1 - \frac{\gamma_2 x}{1 - \dots}}}$$

For diagonal weights $b_0, b_1, \dots$ and downward weights $\lambda_1, \lambda_2, \dots$, write

$$J(b, \lambda; x) = \frac{1}{1 - b_0 x - \frac{\lambda_1 x^2}{1 - b_1 x - \frac{\lambda_2 x^2}{1 - b_2 x - \dots}}}$$

These definitions make sense over any commutative coefficient ring. At any given degree only finitely many levels can contribute. The elementary contraction and translation rules are classical [6]; we record their exact indexing.

Lemma 3.1. *The following identities hold formally.*

(i) The even contraction of $S(\gamma; x)$ has

$$b_0 = \gamma_1, \quad b_k = \gamma_{2k} + \gamma_{2k+1} \quad (k \geq 1), \quad \lambda_k = \gamma_{2k-1}\gamma_{2k}.$$

(ii) If $S = S(\gamma; x)$, then $(S - 1)/(\gamma_1 x)$, whenever the notation denotes the corresponding formal tail product, has J-fraction

$$b_k = \gamma_{2k+1} + \gamma_{2k+2}, \quad \lambda_k = \gamma_{2k}\gamma_{2k+1}.$$

For a numerical nonzero γ_1 this is ordinary division.

(iii) If $F = J(b, \lambda; x)$, then $(1 - cx)^{-1}F(x/(1 - cx)) = J(b + c, \lambda; x)$.

Proof. Let $S_j = (1 - \gamma_{j+1}xS_{j+1})^{-1}$ be the S-fraction tails. For $V_j = S_jS_{j+1}$, direct multiplication gives

$$V_j^{-1} = 1 - (\gamma_{j+1} + \gamma_{j+2})x - \gamma_{j+2}\gamma_{j+3}x^2V_{j+2}.$$

Indeed $S_{j+2} - 1 = \gamma_{j+3}xS_{j+2}S_{j+3}$. Also $(S_0 - 1)/(\gamma_1 x) = S_0S_1 = V_0$, proving (ii). For (i), rewrite $S_0^{-1} = 1 - \gamma_1x - \gamma_1\gamma_2x^2V_1$ and use the same recursion. For (iii), substitute $x/(1 - cx)$ in a finite J-fraction and successively multiply each denominator by $1 - cx$. The coefficient of every fixed power of x then stabilizes as the finite depth increases. $\square$

3.2 A self-contained route to the tangent fraction

Lemma 3.2. *The tangent moment generating series satisfies*

$$\sum_{n \geq 0} T_n x^n = S(\gamma; x), \quad \gamma_j = j(j+1).$$

Proof. Put $f_k(z) = \tan^k z \sec^2 z$, for $k \geq 0$. A differentiation and the identity $\sec^2 z = 1 + \tan^2 z$ give (with the first term omitted when $k = 0$)

$$Df_k = kf_{k-1} + (k+2)f_{k+1}, \quad f_k(0) = \mathbf{1}_{k=0}.$$

Thus $D^{2n}f_0(0) = T_n$ is a sum over walks of length $2n$ that start and end at height zero, stay nonnegative, and have steps ± 1 . An upward crossing from height $j - 1$ has weight $j + 1$, and its downward crossing has weight j . A returning walk crosses each level equally often in the two directions. Its weight therefore equals that of a Dyck path whose up steps have weight one and whose down step from height j has weight $j(j + 1)$. First-return decomposition at each height gives $S_j = 1 + \gamma_{j+1}xS_{j+1}S_j$, the defining recursion of the S-fraction. $\square$

Theorem 3.3 (Bala's displayed conjecture). *Let $A(x) = \sum_{n \geq 0} a_n x^n$. Define*

$$\beta_{2j-1} = (2j+1)^2, \quad \beta_{2j} = 4j(j+1) \quad (j \geq 1). \quad (8)$$

Then, in $\mathbb{Z}[[x]]$,

$$A(x) = \frac{1 + 2xS(\beta; x)}{1 - x}. \quad (9)$$

The successive weights inside $S(\beta; x)$ are 9, 8, 25, 24, 49, 48, 81, 80, ...

Proof. Let $T(x) = \sum T_n x^n$ and $V(x) = (T(x) - 1)/(2x) = \sum T_{n+1} x^n / 2$. Lemma 3.1(ii) applied to Lemma 3.2 gives the J-fraction of V :

$$b_k = 8(k+1)^2, \quad \lambda_k = 4k(k+1)(2k+1)^2. \quad (10)$$

The binomial transform identity is $A(x) = (1-x)^{-1}T(x/(1-x))$. Consequently

$$B(x) := \frac{(1-x)A(x) - 1}{2x} = \frac{1}{1-x}V\left(\frac{x}{1-x}\right).$$

Its J-diagonals are $8(k+1)^2 + 1$, with the same λ_k . On the other hand, the even contraction of (8) has $b_0 = 9$ and, for $k \geq 1$,

$$\beta_{2k} + \beta_{2k+1} = 8(k+1)^2 + 1, \quad \beta_{2k-1}\beta_{2k} = 4k(k+1)(2k+1)^2.$$

Thus the two J-fractions coincide coefficientwise. Hence $B = S(\beta; x)$, which rearranges to (9). $\square$

Remark 3.4. The result is an application of classical contraction and binomial-shift identities, not a new theory of continued fractions. Its value here is that it proves exactly the later conjectural formula, with no finite-depth assumption or experimental extrapolation.

4 The valuation observation and finite modular structure

Even contraction of the tangent fraction and translation by c give

$$\sum_{n \geq 0} A_n(c) x^n = J(b, \lambda; x), \quad b_k = 8k^2 + 8k + 2 + c, \quad \lambda_k = 4k^2(4k^2 - 1). \quad (11)$$

This also identifies $A_n(c)$ as weighted Motzkin path sums: horizontal steps at height k have weight b_k , upward steps weight one, and downward steps from height k weight λ_k . First-return decomposition proves this interpretation directly.

Theorem 4.1 (An exact congruence). *For every $n \geq 1$, $a_n \equiv 3 \pmod{9}$. In particular, $v_3(a_n) = 1$ for every $n \geq 1$.*

Proof. At $c = 1$ the relevant initial weights are

$$b_0 = 3, \quad b_1 = 19, \quad \lambda_1 = 12, \quad \lambda_2 = 240.$$

Every Motzkin path reaching height two contains the factor $\lambda_1 \lambda_2 = 2880$, divisible by nine. Every path with two or more excursions from height zero contains λ_1^2 , also divisible by nine. A path with one such excursion and at least one horizontal step at height zero has a factor $\lambda_1 b_0$, again divisible by nine. For $n \geq 2$ the path consisting only of zero-height horizontal steps has weight $3^n \equiv 0 \pmod{9}$. The only remaining type is a single upward step, then $n-2$ horizontal steps at height one, then a downward step. Its weight is $12 \cdot 19^{n-2} \equiv 3 \pmod{9}$. For $n = 1$ the weight is $b_0 = 3$. This proves the congruence at every index. $\square$

Corollary 4.2. *For every positive integer m and integer c , the formal series $\sum A_n(c) x^n$ is rational over $\mathbb{Z}/m\mathbb{Z}$. Its coefficient sequence modulo m is eventually periodic.*

Proof. The weight $\lambda_m = 4m^2(4m^2 - 1)$ vanishes modulo m . Paths reaching height m therefore vanish, so the series is the finite J-fraction with heights $0, \dots, m - 1$. Its denominator has constant term one. The coefficients obey a finite linear recurrence over a finite ring; the finite vector of successive coefficients evolves deterministically through a finite state space and is eventually periodic. $\square$

The last corollary does not determine a minimal period. It is a structural consequence that can support later, more refined congruence calculations. For the particular modulus nine, Theorem 4.1 says simply $A(x) \equiv 1 + 3x/(1 - x) \pmod{9}$.

5 Positive measures, uniqueness, and Hankel determinants

Proposition 5.1 (Exact moment representation). *For every integer $n \geq 0$ and real $c \geq 0$,*

$$T_n = \int_0^\infty y^{2n} \frac{y}{\sinh(\alpha y)} dy, \quad (12)$$

$$A_n(c) = \int_0^\infty (c + y^2)^n \frac{y}{\sinh(\alpha y)} dy. \quad (13)$$

In particular, $y/\sinh(\alpha y)$ is a probability density on $[0, \infty)$. The corresponding Stieltjes measure for $A_n(c)$ is

$$d\mu_c(t) = \frac{\mathbf{1}_{t > c}}{2 \sinh(\alpha \sqrt{t - c})} dt. \quad (14)$$

Proof. The classical cosine product [7] and its logarithmic derivative give, for $|z| < \alpha$,

$$\tan z = \sum_{\substack{q \geq 1 \\ q \text{ odd}}} \frac{2z}{\alpha^2 q^2 - z^2}.$$

Comparison of Taylor coefficients yields

$$T_n = 2\Gamma(2n + 2)\alpha^{-2n-2} \sum_{q \text{ odd} \geq 1} q^{-2n-2}. \quad (15)$$

On the positive real axis, $1/\sinh(\alpha y) = 2 \sum_{q \text{ odd} \geq 1} e^{-\alpha q y}$. Termwise integration is justified by nonnegativity and gives (12). The binomial theorem gives (13). The case $n = 0$ has value $T_0 = 1$. Finally, set $t = c + y^2$. $\square$

The endpoint density in (14) has an integrable $(t - c)^{-1/2}$ singularity. Its large- t tail is exponential in $\sqrt{t}$, not in t ; this is why the ordinary moment generating series has zero radius but a useful positive integral representation.

Proposition 5.2 (Moment determinacy). *For fixed $c \geq 0$, the sequence $(A_n(c))_{n \geq 0}$ has exactly one representing probability measure on $[0, \infty)$.*

Proof. There are constants $C, D > 0$, depending on c , such that $A_n(c) \leq CD^{2n}(2n + 1)!$. For example, $(c + y^2)^n \leq 2^n(c^n + y^{2n})$ and (15) give such a bound. For any representing measure ν , replace t by the symmetric pair $\pm\sqrt{t}$ to obtain a symmetric probability measure $\tilde{\nu}$ on $\mathbb{R}$. Its even moments are $A_n(c)$ and its odd moments are zero. For sufficiently small $\delta > 0$, monotone convergence gives

$$\int \cosh(\delta x) d\tilde{\nu}(x) = \sum_{n \geq 0} \frac{\delta^{2n} A_n(c)}{(2n)!} < \infty.$$

Therefore its moment generating function is analytic in a neighborhood of zero and is determined by its moments. Two such symmetric measures have the same moment generating function there, hence the same characteristic function and the same measure. Pushing forward by $x \mapsto x^2$ proves uniqueness of ν . $\square$

Proposition 5.3 (Hankel products). *For $N \geq 1$,*

$$\det(A_{i+j}(c))_{0 \leq i, j < N} = \prod_{k=1}^{N-1} [4k^2(4k^2 - 1)]^{N-k}. \quad (16)$$

If $B_n(c) = \frac{1}{2} \sum_{k=0}^n \binom{n}{k} c^{n-k} T_{k+1}$, then

$$\det(B_{i+j}(c))_{0 \leq i, j < N} = \prod_{k=1}^{N-1} [4k(k+1)(2k+1)^2]^{N-k}. \quad (17)$$

Both determinants are independent of c and strictly positive.

Proof. Here is an algebraic justification of the usual J-fraction determinant rule. On the monic polynomial basis $p_0 = 1, p_1, p_2, \dots$, let multiplication by the variable have recurrence

$$tp_k = p_{k+1} + b_k p_k + \lambda_k p_{k-1}, \quad \lambda_0 p_{-1} = 0.$$

Declare this basis orthogonal, with squared norms $h_0 = 1$ and $h_k = \lambda_1 \cdots \lambda_k$. The recurrence makes multiplication by t symmetric for this bilinear form. Its moments $\langle 1, t^n 1 \rangle$ are exactly the weighted Motzkin path sums and hence the coefficients of the J-fraction. The change from monomials to monic p_k is unit triangular, so the Hankel determinant is $\prod_{k=0}^{N-1} h_k$. Apply this to (11) and to the translated version of (10). The formulas follow without any analytic convergence assumption on the generating functions. $\square$

6 Exact exponential sectors and all-order fixed-shift expansions

Let

$$F_n = 2\Gamma(2n+2)\alpha^{-2n-2}, \quad H_n(z) = \sum_{j=0}^n \frac{(z/4)^j}{j!(n+\frac{1}{2})_j}, \quad (18)$$

where $(u)_0 = 1$. The smallest factor in a nonempty denominator of H_n is at least $3/2$, so no zero denominator occurs.

Theorem 6.1 (Exact sectors and an explicit tail bound). *For every $n \geq 0$ and $c \geq 0$,*

$$\frac{A_n(c)}{F_n} = \sum_{q \text{ odd} \geq 1} q^{-2n-2} H_n(c\alpha^2 q^2). \quad (19)$$

The series is convergent and all its terms are nonnegative. If $Q \geq 1$ is odd and $R_{n,Q}(c)$ is the contribution to $A_n(c)$ from $q \geq Q$, then

$$0 \leq \frac{R_{n,Q}(c)}{F_n} \leq \left(1 + \frac{Q}{2}\right) Q^{-2n-2} H_n(c\alpha^2 Q^2). \quad (20)$$

Proof. Insert (15) into the binomial transform and put $j = n - k$. The duplication identity, or cancellation of consecutive factorials, gives

$$\binom{n}{j} \frac{\Gamma(2n - 2j + 2)}{\Gamma(2n + 2)} = \frac{1}{4^j j! (n + \frac{1}{2})_j}.$$

After factoring q^{-2n-2} , this is exactly (19). At fixed n the last polynomial term decays as q^{-2} ; every other term decays faster, proving convergence. For every real $r \geq 2$, monotonicity and an integral comparison give

$$\sum_{j \geq 0} (Q + 2j)^{-r} \leq Q^{-r} + \frac{1}{2} \int_Q^\infty u^{-r} du = Q^{-r} \left(1 + \frac{Q}{2(r-1)} \right) \leq Q^{-r} (1 + Q/2).$$

Apply this separately to each nonnegative coefficient of H_n ; its corresponding exponent is $r = 2n + 2 - 2j \geq 2$. $\square$

Theorem 6.2 (A coefficient algorithm to every algebraic order). *Uniformly for z in each fixed compact subset of $\mathbb{C}$,*

$$H_n(z) \sim \sum_{r \geq 0} \frac{C_r(z)}{n^r}, \quad (21)$$

where

$$C_r(z) = \sum_{j=0}^r \frac{(z/4)^j}{j!} [t^{r-j}] \prod_{\ell=0}^{j-1} (1 + (\tfrac{1}{2} - \ell)t)^{-1}. \quad (22)$$

In particular,

$$\begin{aligned} C_0(z) &= 1, & C_1(z) &= \frac{z}{4}, \\ C_2(z) &= -\frac{z}{8} + \frac{z^2}{32}, & C_3(z) &= \frac{z}{16} + \frac{z^3}{384}, \\ C_4(z) &= -\frac{z}{32} + \frac{z^2}{128} + \frac{z^3}{256} + \frac{z^4}{6144}. \end{aligned}$$

Proof. For a fixed j the summand is

$$\frac{(z/4)^j}{j! n^j} \prod_{\ell=0}^{j-1} \left(1 + \frac{1/2 - \ell}{n} \right)^{-1}.$$

Taylor expansion of the finite product gives (22). It remains to justify discarding unbounded j . For $j \leq n/2$, every falling-factorial factor is at least $n/2$, so the absolute value of the summand is at most $(|z|/(2n))^j/j!$. The sum of these bounds over $j \geq M$ is $O(n^{-M})$, uniformly for bounded z . For $j > n/2$, at least $\lfloor n/2 \rfloor$ factors are at least $n/2$ and all remaining factors exceed one. Summing the numerator divided by $j!$ bounds this part by $e^{|z|/4} (n/2)^{-\lfloor n/2 \rfloor}$. The finite terms with $j < M$ have uniform Taylor remainders, completing the proof at each prescribed order. $\square$

Combining these two theorems gives an explicit description of every fixed odd-pole sector. For a fixed odd cutoff Q and fixed c , the omitted sectors are $O(F_n Q^{-2n-2})$. This statement uses the exact H_n at the cutoff, not a claim uniform in arbitrarily growing Q after replacing H_n by a fixed- z expansion.

6.1 Why the exact sectors matter

The first exponentially small correction is meaningful in the form

$$\frac{A_n(c)}{F_n H_n(c\alpha^2)} = 1 + 3^{-2n-2} \frac{H_n(9c\alpha^2)}{H_n(c\alpha^2)} + O_c(5^{-2n-2}). \quad (23)$$

The exact denominator is essential. Replacing it by a fixed finite inverse-power approximation introduces a polynomial error that is much larger than 3^{-2n} . Formula (23) avoids that error-order confusion and supplies a rigorous positive residual bound through (20).

6.2 The logarithmic expansion and the prior leading equivalent

Write $\log(\sum_{r \geq 0} C_r(z)t^r) = \sum_{r \geq 1} \ell_r(z)t^r$. The first six logarithmic coefficients, generated by the supplied code, are

$$\begin{aligned} \ell_1(z) &= \frac{z}{4} \\ \ell_2(z) &= -\frac{z}{8} \\ \ell_3(z) &= \frac{z^2}{32} + \frac{z}{16} \\ \ell_4(z) &= -\frac{z^2}{64} - \frac{z}{32} \\ \ell_5(z) &= \frac{z^3}{96} + \frac{z^2}{64} + \frac{z}{64} \\ \ell_6(z) &= \frac{z^3}{192} - \frac{z^2}{128} - \frac{z}{128} \end{aligned}$$

For fixed $c \geq 0$, Stirling's expansion [8] and the exponentially small sectors give

$$\log A_n(c) \sim 2n \log \frac{2n}{\alpha e} + \frac{3}{2} \log n + \log \frac{8\sqrt{\pi}}{\alpha^2} + \sum_{r \geq 1} \frac{h_r(c)}{n^r}. \quad (24)$$

The coefficient rule is

$$h_r(c) = \ell_r(c\alpha^2) + \frac{(-1)^{r+1}}{r2^r} + \begin{cases} \frac{B_{r+1}}{(r+1)r2^r}, & r \text{ odd,} \\ 0, & r \text{ even,} \end{cases} \quad (25)$$

where B_j are Bernoulli numbers. For example,

$$\begin{aligned} h_1(c) &= \frac{13}{24} + \frac{c\alpha^2}{4}, \\ h_2(c) &= -\frac{1}{8} - \frac{c\alpha^2}{8}, \\ h_3(c) &= \frac{119}{2880} + \frac{c\alpha^2}{16} + \frac{c^2\alpha^4}{32}. \end{aligned} \quad (26)$$

One way to verify the constants is to use $\Gamma(2n+2) = (2n+1)\Gamma(2n+1)$: the extra logarithm is $\log(1 + 1/(2n))$, and the remaining correction is the ordinary Stirling series at $2n$.

At $c = 1$, exponentiating the leading part gives

$$a_n \sim \frac{2^{4n+5} n^{2n+3/2}}{e^{2n} \pi^{2n+3/2}}, \quad (27)$$

which is the leading equivalent already recorded by Kotesovec in 2015 [1]. Formula (27) is therefore a consistency check, not a newly solved asymptotic problem. The all-order and exponentially resolved refinements above go beyond that leading formula in the inspected entry.

7 Quadratic shifts: the geometry of a first-order transition

The fixed- c expansion cannot be inserted blindly when c grows like n^2 . In that regime an entirely different contribution competes with the factorial tail. Set $c = \tau n^2$ in (13), then put $y = nu$:

$$A_n(\tau n^2) = n^{2n+2} \int_0^\infty (\tau + u^2)^n \frac{u}{\sinh(\alpha n u)} du. \quad (28)$$

Near a fixed positive u , the hyperbolic factor is asymptotic to $2e^{-\alpha n u}$; the phase is

$$\phi_\tau(u) = \log(\tau + u^2) - \alpha u. \quad (29)$$

The region $y = O(1)$, equivalently $u = O(1/n)$, is an endpoint contribution with leading value $(\tau n^2)^n$. It must not be estimated by replacing $1/\sinh(\alpha y)$ by its large- y first exponential.

For $0 < \tau < \alpha^{-2}$ the stationary points are

$$u_\pm = \frac{1 \pm \sqrt{1 - \alpha^2 \tau}}{\alpha}.$$

The larger one is a local maximum. Parameterize it by

$$s = \alpha u_+ = 1 + \sqrt{1 - \alpha^2 \tau}, \quad \tau = \frac{s(2-s)}{\alpha^2}, \quad 1 < s < 2. \quad (30)$$

The saddle's logarithmic advantage over the endpoint is

$$\Delta(\tau) = \phi_\tau(u_+) - \log \tau = \log \frac{2}{2-s} - s. \quad (31)$$

As a function of s , its derivative is $(s-1)/(2-s) > 0$. It is negative at $s = 1$ and tends to infinity at $s = 2$. Thus there is a unique s_* in $(1, 2)$ solving (4).

The curvature, amplitude, and derivative of the phase difference are

$$\kappa(\tau) = -\phi''_\tau(u_+) = \alpha^2 \frac{s-1}{s}, \quad (32)$$

$$K(\tau) = 2u_+ \sqrt{\frac{2\pi}{\kappa(\tau)}} = \frac{2\sqrt{2\pi} s^{3/2}}{\alpha^2 \sqrt{s-1}}, \quad (33)$$

$$-\Delta'(\tau) = \frac{\alpha^2}{2(2-s)}. \quad (34)$$

In particular, with $d_* = -\Delta'(\tau_*)$ and $K_* = K(\tau_*)$,

$$\begin{aligned} s_* &= 1.5936242600400400923 \dots, \\ \tau_* &= 0.2624665433684190267 \dots, \\ d_* &= 3.0358617132452985767 \dots, \\ K_* &= 5.3052182371146210849 \dots \end{aligned}$$

These displayed digits are numerical evaluations of exact definitions, not interval-certified decimal bounds.

[Write note, batch 7301, 1 October 2026.] Notation across reports: $s_* = 2 + W_0(-2e^{-2}) = 1.59362426\dots$ is the constant written τ in the A277364 report [15]; here τ is the shift ratio c/n^2 , and $\tau_* = s_*(2 - s_*)/\alpha^2 = 0.26246654\dots$ is a different number. Do not read the A277364 report's τ as this report's τ_* . The shared equation $2 - s = 2e^{-s}$ arises there from the omitted tail of a Stirling sum and here from the endpoint–saddle balance $\Delta(\tau) = 0$; neither report uses the other's argument.

Proposition 7.1 (Fixed-parameter regimes). *If $0 < \tau < \tau_*$ is fixed, then*

$$A_n(\tau n^2) \sim n^{2n} e^{n\phi_\tau(u_+)} K(\tau) n^{3/2}.$$

If $\tau > \tau_$ is fixed, then $A_n(\tau n^2) \sim (\tau n^2)^n$. At $\tau = \tau_*$ the saddle, not the endpoint, still dominates by the factor $K_* n^{3/2}$.*

Proof. When the larger stationary point exists, comparison of the endpoint value $\log \tau$ and the saddle value is exactly the sign of Δ . There are no other local maxima. For $\tau \geq \alpha^{-2}$ the phase has no larger interior value, so the endpoint is the global maximum. The endpoint integral tends to one after division by $(\tau n^2)^n$; the saddle has amplitude $2n^2 u_+$ and Gaussian width $n^{-1/2}$ in u , giving the factor in (33). The localization argument in the next section supplies uniform errors and proves these leading assertions. $\square$

The terminology “first-order” refers to the discontinuous limiting fraction of the binomial sum's active index, proved in Section 9. It is not merely a change in a prefactor.

8 A uniform two-contribution expansion to all orders

Choose a sufficiently small closed interval I about τ_* contained in $(0, \alpha^{-2})$. All constants implicit in the following theorem may depend on I and the fixed truncation order, but not on $\tau \in I$. For a list $v_1, \dots, v_r$, let $e_j(v_1, \dots, v_r)$ denote its elementary symmetric polynomial, with $e_0 = 1$ and $e_j = 0$ outside its natural range.

Theorem 8.1 (Uniform all-order endpoint–saddle expansion). *For every fixed integer $M \geq 1$, uniformly for $\tau \in I$,*

$$\begin{aligned} \frac{A_n(\tau n^2)}{(\tau n^2)^n} &= \sum_{m=0}^{M-1} \frac{E_m(\tau)}{n^m} + K(\tau) n^{3/2} e^{n\Delta(\tau)} \sum_{m=0}^{M-1} \frac{L_m(\tau)}{n^m} \\ &\quad + O_{I,M}\left(n^{-M} \left[1 + n^{3/2} e^{n\Delta(\tau)}\right]\right). \end{aligned} \quad (35)$$

Here $E_0 = L_0 = 1$ and

$$E_m(\tau) = \sum_{k=0}^m \frac{T_k}{k! \tau^k} (-1)^{m-k} e_{m-k}(0, 1, \dots, k-1). \quad (36)$$

The saddle coefficients have the finite Gaussian coefficient rule

$$L_m(\tau) = [\varepsilon^{2m}] \mathbb{E} \left[\left(1 + \frac{\varepsilon Z}{u_+}\right) \exp \left\{ \sum_{r \geq 3} \frac{\phi_\tau^{(r)}(u_+)}{r!} Z^r \varepsilon^{r-2} \right\} \right], \quad (37)$$

where $Z \sim \mathcal{N}(0, \kappa(\tau)^{-1})$ and the expectation is applied to each polynomial coefficient. In particular,

$$E_1(\tau) = \frac{2}{\tau}, \quad E_2(\tau) = \frac{8}{\tau^2}, \quad E_3(\tau) = \frac{136}{3\tau^3} - \frac{8}{\tau^2}, \quad (38)$$

$$L_1(\tau) = \frac{2s^3 + 18s^2 - 60s + 45}{24(s-1)^3}. \quad (39)$$

Proof. We give the localization and remainder details, because the endpoint and the interior saddle live on different scales.

Endpoint. Fix a sufficiently small $\eta > 0$, uniformly over $\tau \in I$, so that $\eta/\min I < \alpha/2$ and η lies below the smaller stationary point throughout I . On $0 \leq y \leq \eta n$,

$$\left(1 + \frac{y^2}{\tau n^2}\right)^n \leq \exp\left(\frac{y^2}{\tau n}\right) \leq \exp\left(\frac{\eta y}{\min I}\right).$$

Multiplication by $y/\sinh(\alpha y)$ therefore leaves an integrable exponential majorant, uniformly in n and τ . At each prescribed order, expand

$$\left(1 + \frac{y^2}{\tau n^2}\right)^n = \sum_{k=0}^n \frac{(n)_k}{k!} \frac{y^{2k}}{\tau^k n^{2k}}.$$

For a fixed coefficient of n^{-m} , only $k \leq m$ can contribute; expanding $(n)_k = n^k \prod_{j=0}^{k-1} (1 - j/n)$ gives precisely (36) after integration.

For a fully uniform remainder bound, introduce the analytic function

$$G_y(t) = \exp\left(\frac{\log(1 + y^2 t^2 / \tau)}{t}\right), \quad G_y(0) = 1.$$

Choose $\rho > 0$ so small that $\rho^2/\min I < 1/2$ and $2\rho/\min I < \alpha/2$. On the complex circle $|t| = \rho/(1+y)$ the logarithm is its branch near one and $|G_y(t)| \leq \exp(2\rho y/\min I)$. Reduce η to at most $\rho/4$. For all sufficiently large n and $0 \leq y \leq \eta n$, the evaluation point $t = 1/n$ is inside half this circle. Cauchy's coefficient estimate bounds the Taylor remainder after t^{M-1} by

$$2 \exp(2\rho y/\min I) \left(\frac{1+y}{\rho n}\right)^M.$$

Its product with $y/\sinh(\alpha y)$ is integrable uniformly in n and τ , proving the required $O(n^{-M})$ integrated error. The Taylor coefficients agree with the finite formal calculation above. Extending their integrals from ηn to infinity changes only an exponentially small amount.

Interior saddle. Choose a small neighborhood of $u_+(\tau)$ whose distance from zero is uniformly positive and whose curvature is bounded away from zero. In this region

$$\frac{1}{\sinh(\alpha n u)} = 2e^{-\alpha n u} (1 + O(e^{-\rho n}))$$

for some $\rho > 0$. Set $u = u_+ + n^{-1/2}z$ in the first exponential term of (28). The quadratic term gives the centered Gaussian with variance κ^{-1} , and the amplitude u/u_+ is $1 + n^{-1/2}z/u_+$. Taylor's formula for the remaining phase gives (37). Every coefficient involves only finitely many derivatives and Gaussian moments. Odd powers of $n^{-1/2}$ integrate to zero. The standard localized Laplace

estimate, obtained by restricting first to $|z| \leq n^\delta$ and then using Gaussian domination on its complement, gives relative error $O(n^{-M})$ uniformly on I [9]. The prefactor is $K(\tau)n^{3/2}e^{n\Delta(\tau)}$.

Complement. Outside the endpoint region and the saddle neighborhood, $\phi_\tau(u)$ is less than $\max\{\log \tau, \phi_\tau(u_+)\}$ by a fixed positive amount on each sufficiently large compact u interval. For large u , the term $-\alpha u$ dominates $\log(\tau + u^2)$ uniformly, giving the same type of exponential bound on the unbounded tail. The compact interval I may be reduced so that these gaps remain uniform. Thus this complement is exponentially small relative to the larger of the two contributions, and is absorbed by the remainder in (35).

Finally, evaluating (36) yields (38). For the first saddle correction, Gaussian moments give

$$L_1 = \frac{\phi'''(u_+)}{2u_+\kappa^2} + \frac{\phi''''(u_+)}{8\kappa^2} + \frac{5\phi'''(u_+)^2}{24\kappa^3}.$$

Substitution of (30) and (32) simplifies to (39). $\square$

Remark 8.2 (Scope of the uniform expansion). This theorem is uniform near the coexistence point, not near the coalescing stationary points at $\tau = \alpha^{-2}$. It also does not make a fixed- c expansion uniform up to $c \asymp n^2$. The two asymptotic problems use different localizations. If one of the terms in (35) is exponentially smaller than the other, a finite inverse-power error in the larger term can mask it; the underlying localized integrals still have their separate expansions.

9 The moving window and the probability of a macroscopic index

Theorem 9.1 (Finite-size coexistence window). *For fixed bounded real w , define*

$$\tau_n(w) = \tau_* + \frac{\frac{3}{2} \log n + \log K_* + w}{d_* n}. \quad (40)$$

Then, uniformly for w in compact real sets,

$$\frac{A_n(\tau_n(w)n^2)}{(\tau_n(w)n^2)^n} = 1 + e^{-w} + O\left(\frac{(\log n)^2}{n}\right). \quad (41)$$

Proof. Write $\delta_n = \tau_n - \tau_*$. Taylor expansion gives $n\Delta(\tau_n) = -d_* n\delta_n + O(n\delta_n^2)$ and $\log K(\tau_n) = \log K_* + O(\delta_n)$. Since $\delta_n = O((\log n)/n)$,

$$\log(K(\tau_n)n^{3/2}e^{n\Delta(\tau_n)}) = -w + O((\log n)^2/n).$$

The endpoint factor is $1 + O(1/n)$ and the saddle relative correction is $1 + O(1/n)$ by Theorem 8.1. Combining them proves the result. $\square$

A useful more accurate centering is the unique nearby solution of

$$n\Delta(\tau) + \frac{3}{2} \log n + \log K(\tau) = -w. \quad (42)$$

Its derivative with respect to τ is $-d_* n + O(\log n)$, so existence and uniqueness in the relevant neighborhood follow for large n . Under this centering the leading saddle factor is exactly e^{-w} and the error in (41) improves to $O(1/n)$. Using the coefficients E_m, L_m in the balance equation gives successive higher-order corrections. Equation (40) is the simple explicit first approximation, not the most accurate numerical prescription.

9.1 A positive random-index representation

For $c > 0$ define a random index $K_n \in \{0, \dots, n\}$ by

$$\mathbb{P}_c(K_n = k) = \frac{\binom{n}{k} c^{n-k} T_k}{A_n(c)}. \quad (43)$$

Under the probability measure on $y \geq 0$ proportional to $(c + y^2)^n y / \sinh(\alpha y)$, conditionally take

$$K_n | y \sim \text{Bin}\left(n, \frac{y^2}{c + y^2}\right). \quad (44)$$

Integrating the conditional mass function recovers (43), so this is an exact mixture construction, not an approximation.

Theorem 9.2 (Discontinuous fraction and critical mixture). *If $0 < \tau < \tau_*$ is fixed and $c = \tau n^2$, then $K_n/n \rightarrow s(\tau)/2$ in probability. If $\tau > \tau_*$ is fixed, then $K_n \rightarrow 0$ in probability. In the moving window (40),*

$$\frac{K_n}{n} \Rightarrow \frac{1}{1 + e^{-w}} \delta_0 + \frac{e^{-w}}{1 + e^{-w}} \delta_{s_*/2}. \quad (45)$$

The macroscopic location at coexistence is $s_*/2 = 0.796812130020020\dots$

Proof. The endpoint tilted measure is concentrated on $y = O(1)$, while the saddle measure has $y/n \rightarrow u_+$. Conditional binomial variance divided by n^2 is at most $1/(4n)$. The conditional mean fraction tends to zero in the endpoint region and, in the saddle region, to

$$\frac{u_+^2}{\tau + u_+^2} = \frac{s}{2}.$$

The two relative masses follow from (35) and (41). In the fixed endpoint-dominated regime the stronger claim follows directly from $\mathbb{P}(K_n = 0) = c^n/A_n(c) \rightarrow 1$. $\square$

Theorem 9.3 (Gaussian fluctuations within the macroscopic component). *For fixed $0 < \tau < \tau_*$,*

$$\frac{K_n - ns(\tau)/2}{\sqrt{n}} \Rightarrow \mathcal{N}(0, v(\tau)), \quad v(\tau) = \frac{s(2-s)}{4(s-1)}. \quad (46)$$

In the critical window the same conclusion holds conditionally on the saddle component, with $s = s(\tau_n)$ and variance tending to $v(\tau_)$. Equivalently, one may condition on $K_n/n > \varepsilon$ for any fixed $0 < \varepsilon < s_*/2$.*

Proof. Localized Laplace expansion gives $\sqrt{n}(y/n - u_+) \Rightarrow \mathcal{N}(0, \kappa^{-1})$. Let $p(u) = u^2/(\tau + u^2)$. Uniformly near u_+ , the conditional binomial characteristic function for its centered fluctuations converges to that of $\mathcal{N}(0, p(u_+)(1 - p(u_+)))$. At the $\sqrt{n}$ scale the variation in its conditional mean is $p'(u_+)\sqrt{n}(y/n - u_+)$. The limiting characteristic function is thus the product of the two Gaussian factors, with variance

$$p(u_+)(1 - p(u_+)) + \frac{p'(u_+)^2}{\kappa}.$$

Since $p(u_+) = s/2$ and $p'(u_+) = \alpha(2 - s)/2$, simplification gives (46). Endpoint leakage under the stated conditioning tends to zero by the separated concentration regions. The same localization and conditional characteristic-function argument is uniform for $\tau_n \rightarrow \tau_*$. $\square$

The critical limiting law is a two-point law at scale n , with a Gaussian width of order $\sqrt{n}$ around its nonzero point. A single Gaussian approximation to the entire distribution would therefore be incorrect.

10 A universality theorem for exponentially decaying moment tails

In this section α is an arbitrary positive tail rate; in Section 12 it again denotes $\pi/2$. The logarithmic displacement is not specific to tangent numbers. The following result identifies the parameters on which it depends. It is a consequence of a competition between an unscaled probability-mass contribution and a scaled interior Laplace saddle; it does not assert a new general method of asymptotic integration.

Theorem 10.1 (Universal shifted-moment transition). *Let μ be a probability measure on $[0, \infty)$ whose restriction beyond some finite point has a density*

$$w(y) = Cy^\beta e^{-\alpha y}(1 + O(y^{-1})), \quad C, \alpha > 0, \quad \beta \in \mathbb{R}. \quad (47)$$

Fix $p > 1$ and put

$$m_k = \int y^{pk} d\mu(y), \quad M_n(c) = \sum_{k=0}^n \binom{n}{k} c^{n-k} m_k = \int (c + y^p)^n d\mu(y).$$

There is a unique $s_p \in (p-1, p)$ such that

$$p - s_p = pe^{-s_p}, \quad \tau_{p,*} = \frac{s_p^{p-1}(p - s_p)}{\alpha^p}. \quad (48)$$

Define

$$\kappa_* = \alpha^2 \frac{s_p - p + 1}{s_p}, \quad K_{p,*} = C \left(\frac{s_p}{\alpha} \right)^\beta \sqrt{\frac{2\pi}{\kappa_*}}, \quad (49)$$

$$d_{p,*} = \frac{\alpha^p}{ps_p^{p-2}(p - s_p)}, \quad b = \beta + \frac{1}{2}. \quad (50)$$

For

$$\tau_n(w) = \tau_{p,*} + \frac{b \log n + \log K_{p,*} + w}{d_{p,*}n}, \quad (51)$$

we have, uniformly for bounded real w ,

$$\frac{M_n(\tau_n(w)n^p)}{(\tau_n(w)n^p)^n} = 1 + e^{-w} + O\left(n^{1-p} + \frac{(\log n)^2}{n}\right). \quad (52)$$

The random binomial index with masses proportional to $\binom{n}{k} c^{n-k} m_k$ satisfies

$$K_n/n \implies \frac{1}{1 + e^{-w}} \delta_0 + \frac{e^{-w}}{1 + e^{-w}} \delta_{s_p/p}. \quad (53)$$

For fixed $0 < \tau < \tau_{p,*}$, the centered index $(K_n - ns/p)/\sqrt{n}$ converges to a centered Gaussian with variance

$$v_p(\tau) = \frac{s(p-s)}{p^2(s-p+1)}, \quad \tau = \frac{s^{p-1}(p-s)}{\alpha^p}, \quad p-1 < s < p. \quad (54)$$

The same variance, evaluated at s_p , describes the macroscopic component at coexistence.

Proof. The relevant phase after $y = nu$ is

$$\phi_{\tau,p}(u) = \log(\tau + u^p) - \alpha u.$$

Its positive stationary points satisfy

$$\tau = \frac{s^{p-1}(p-s)}{\alpha^p}, \quad s = \alpha u.$$

The larger stationary point has $p-1 < s < p$ and negative second derivative $-\alpha^2(s-p+1)/s$. There are at most two positive stationary points, because $s^{p-1}(p-s)$ increases up to $p-1$ and then decreases to zero. At the larger point the excess over the endpoint phase is

$$\Delta_p(\tau) = \log \frac{p}{p-s} - s. \quad (55)$$

As a function of $s \in (p-1, p)$, its derivative is $(s-p+1)/(p-s) > 0$. Its value at the left endpoint is $\log p - (p-1) < 0$, while it tends to infinity at the right endpoint. This proves existence and uniqueness in (48). Differentiating the maximized phase with respect to τ gives $-\Delta'_p(\tau) = \alpha^p/[ps^{p-2}(p-s)]$.

Choose a small endpoint region $y \leq \eta n$, uniformly near the transition. For $p > 1$,

$$\left(1 + \frac{y^p}{\tau n^p}\right)^n \leq \exp\left(\frac{\eta^{p-1}y}{\tau}\right).$$

Taking η small enough preserves an integrable exponential majorant against the tail of μ . Moreover $e^v - 1 \leq ve^v$, so integration of this bound shows that the normalized endpoint contribution is $1 + O(n^{1-p})$. Compactly supported mass, including atoms, contributes in the same way and is included in this leading one.

Near the larger saddle, the tail assumption gives the amplitude $Cn^{\beta+1}u^\beta$ times $1 + O(1/n)$, uniformly in a compact positive u interval. The Gaussian width is $n^{-1/2}$, so the normalized saddle contribution is

$$K_p(\tau)n^{\beta+1/2}e^{n\Delta_p(\tau)}(1 + O(1/n)), \quad K_p(\tau) = C(s/\alpha)^\beta \sqrt{\frac{2\pi s}{\alpha^2(s-p+1)}}.$$

The $O(y^{-1})$ error need not be differentiable: a uniform relative bound on the localized integral suffices. The smooth main amplitude has the usual Laplace estimate. The phase gap on the remaining region is handled as in Theorem 8.1. Inserting (51) and Taylor expanding Δ_p and $\log K_p$ proves (52). The error n^{1-p} must be retained when $1 < p < 2$; replacing it by $1/n$ would in general be false.

The exact conditional construction is now $K_n \mid y \sim \text{Bin}(n, y^p/(c+y^p))$. At the saddle the success probability is s/p , so the same concentration argument proves (53). For the Gaussian fluctuations,

$$p_0 = \frac{s}{p}, \quad p'_0 = \frac{\alpha(p-s)}{p}, \quad \kappa = \alpha^2 \frac{s-p+1}{s}.$$

Their variance is $p_0(1-p_0) + (p'_0)^2/\kappa$, which equals (54). The conditional characteristic-function proof used in Theorem 9.3 applies unchanged. $\square$

10.1 What the universality theorem does and does not cover

The transition location depends on p and the exponential rate α , not on the compact part of the measure or the tail power β . The displacement of the finite-size window does depend on β : its logarithmic coefficient is $\beta + 1/2$. This coefficient may be negative, zero, or positive; the theorem requires none of those cases to be excluded. The assumption that μ is a probability measure fixes the endpoint amplitude to one. Without that normalization the mass of the measure must be inserted explicitly in the balance equation.

The theorem gives a leading uniform transition and a bulk central limit law. It does not claim an all-order expansion for an arbitrary $O(y^{-1})$ tail perturbation. An all-order tail expansion, together with appropriate regularity or controlled remainders, would support an all-order extension; the tangent case has that additional structure. The boundary case $p = 1$ is excluded because its interior stationary point coalesces with the endpoint as the critical parameters are approached.

10.2 Examples beyond A125054

The tangent density has $C = 2$, $\beta = 1$ and $p = 2$, recovering the $\frac{3}{2} \log n$ displacement. The positive density $\operatorname{sech}(\pi y/2)$ on $[0, \infty)$ has even moments the secant numbers E_{2k} , OEIS A000364 [4]. For example, integrating its geometric exponential expansion gives $2\Gamma(2k+1)\alpha^{-2k-1} \sum_{j \geq 0} (-1)^j (2j+1)^{-2k-1}$, the usual secant partial-fraction coefficient. In this case $C = 2$, $\beta = 0$, so the same quadratic-shift transition location has a $\frac{1}{2} \log n$ displacement instead.

For a gamma density

$$d\mu(y) = \frac{\alpha^b}{\Gamma(b)} y^{b-1} e^{-\alpha y} dy, \quad b > 0,$$

the moments are explicitly $m_k = \alpha^{-pk} \Gamma(pk+b)/\Gamma(b)$. Here the logarithmic coefficient is $b - 1/2$. Thus the theorem covers factorial subsequences and generalized factorial moment transforms as well as alternating-permutation numbers.

More generally, a tail $Dy^\gamma e^{-ay^r}$ can be reduced by the substitution $v = y^r$ to the theorem whenever the moment power divided by r exceeds one. The shift scale is then $n^{p/r}$, and the transformed tail power is $(\gamma+1)/r - 1$. This is a direct change-of-variable corollary, not a separate new universality hypothesis.

11 Complex zeros from the real transition and positivity

A useful additional consequence requires no separate deformation to a complex saddle. Positive polynomial coefficients provide the complex bounds needed to promote the real limit to an analytic one. We state the result for the general moment model of Theorem 10.1.

Theorem 11.1 (Complex coexistence limit and a zero array). *Under the hypotheses of Theorem 10.1, extend $\tau_n(w)$ in (51) affinely to complex w and define*

$$F_n(w) = \frac{M_n(n^p \tau_n(w))}{(n^p \tau_n(w))^n}.$$

Then $F_n(w) \rightarrow 1 + e^{-w}$ locally uniformly on $\mathbb{C}$. For every fixed integer j , and all sufficiently large n , there is a simple zero $c_{n,j}$ of $M_n(c)$ such that

$$\frac{c_{n,j}}{n^p} = \tau_{p,*} + \frac{(\beta + \frac{1}{2}) \log n + \log K_{p,*} + (2j+1)\pi i}{d_{p,*} n} + o(n^{-1}). \quad (56)$$

More precisely, in any fixed sufficiently small disk about $w_j = (2j+1)\pi i$, the function F_n has exactly one zero, counting multiplicity, for all sufficiently large n ; that zero tends to w_j .

Proof. On each fixed compact set of w , $\tau_n(w) \rightarrow \tau_{p,*} > 0$, so F_n is analytic there for all sufficiently large n . Write $\tau_n(w) = a_n + ib_n$ with $a_n > 0$. For bounded w we have $b_n = O(1/n)$ and

$$|\tau_n(w)| - \tau_n(\Re w) = \frac{b_n^2}{|\tau_n(w)| + a_n} = O(n^{-2}).$$

Therefore $|\tau_n(w)| = \tau_n(r_n)$ for a real number $r_n = \Re w + O(1/n)$, uniformly on the compact set. Since $M_n(c)$ has nonnegative coefficients, the triangle inequality gives

$$|F_n(w)| \leq \frac{M_n(n^p |\tau_n(w)|)}{(n^p |\tau_n(w)|)^n} = F_n(r_n). \quad (57)$$

The real transition theorem bounds the right side uniformly on compact sets. Cauchy's integral formula now gives uniform derivative bounds on smaller compact disks. By equicontinuity and a diagonal subsequence argument, every subsequence has a further subsequence converging locally uniformly to an analytic function. On the real axis that function must be $1 + e^{-w}$, by Theorem 10.1. The identity theorem therefore identifies every such analytic limit with $1 + e^{-w}$. This proves locally uniform convergence of the entire sequence. The analytic tools are the usual Cauchy estimates and uniqueness of analytic continuation; see also [10].

The limit $1 + e^{-w}$ has a simple zero at each w_j and no other zero in a sufficiently small closed disk about w_j . On the boundary circle its modulus has a positive minimum. Uniform convergence and Rouché's theorem then give exactly one zero of F_n in that disk, counting multiplicity. Shrinking the disk proves convergence to w_j ; count one also proves simplicity. The denominator defining F_n is nonzero there, so these are zeros of M_n . Transforming the coordinate back to c gives (56). $\square$

Corollary 11.2 (Central Poupard polynomial zeros). *For each fixed $j \in \mathbb{Z}$, $A_n(c)$ has a simple zero with*

$$\frac{c_{n,j}}{n^2} = \tau_* + \frac{\frac{3}{2} \log n + \log K_* + (2j+1)\pi i}{d_* n} + o(n^{-1}). \quad (58)$$

In particular, for all sufficiently large n , some zeros of $A_n(c)$ lie in the right half-plane.

Proof. Use $p = 2$, $\beta = 1$, $C = 2$, $\alpha = \pi/2$ in the theorem. The real part of $c_{n,j}/n^2$ tends to $\tau_* > 0$. $\square$

The theorem controls every fixed finite collection of these zeros, not indices j growing with n or the global zero distribution. Its locally uniform complex convergence is qualitative; the stated real-axis error does not automatically become an identical complex error bound. Higher-order complex zero locations require an additional quantitative argument.

12 Inverting the central Poupard growth

12.1 A canonical real interpolation

For real $x \geq 0$ define

$$A_x = \int_0^\infty (1+y^2)^x \frac{y}{\sinh(\alpha y)} dy. \quad (59)$$

This agrees with a_n at integer $x = n$, is strictly increasing, starts at $A_0 = 1$, and tends to infinity. It therefore has a unique inverse $x = x(Y)$ for $Y \geq 1$. The integral is in fact entire in the complex parameter x , since on compact sets of x its absolute value is bounded by a fixed power of $1 + y^2$ times an integrable exponential tail. Only the positive real inverse is used below.

Lemma 12.1. *The fixed-shift logarithmic expansion (24), with $c = 1$ and n replaced by real $x \rightarrow \infty$, holds to every fixed order.*

Proof. First consider one fixed odd exponential q in the density. Its saddle lies near $y = 2x/(\alpha q)$. Restrict to $y \geq \eta x$ with $\eta > 0$ smaller than the saddle location divided by x . Here the generalized binomial expansion of $(1 + y^{-2})^x$ to any fixed number of terms has a controlled remainder, because $x/y^2 = O(1/x)$. Integrating its j th term gives the factor

$$\frac{(x)_j}{j!} \frac{\Gamma(2x - 2j + 2)}{\Gamma(2x + 2)} = \frac{1}{4^j j! (x + \frac{1}{2})_j}.$$

The omitted lower region is exponentially small relative to the saddle if η is sufficiently small; a bounded y region is smaller still. The same localization bounds the integrated remainder at the desired inverse-power order. Thus the coefficients C_r and then ℓ_r are unchanged for real x . The other odd exponentials are exponentially smaller. Stirling's real expansion then gives (24). $\square$

Theorem 12.2 (Lambert inversion through the first two corrections). *Let $L = \log Y$, let W be the positive real Lambert branch, and set*

$$w = W\left(\frac{L}{\alpha e}\right), \quad x_0 = \frac{L}{2w}, \quad C_0 = \log \frac{8\sqrt{\pi}}{\alpha^2}. \quad (60)$$

Define

$$d_0 = -\frac{\frac{3}{2} \log x_0 + C_0}{2(w + 1)}, \quad (61)$$

$$d_1 = -\frac{d_0^2 + \frac{3}{2}d_0 + 13/24 + \alpha^2/4}{2x_0(w + 1)}. \quad (62)$$

As $Y \rightarrow \infty$,

$$x(Y) = x_0 + d_0 + d_1 + O\left(\frac{1}{x_0^2(w + 1)}\right). \quad (63)$$

In particular, $d_0 \rightarrow -3/4$.

Proof. The defining identity for W [11] gives $\log(2x_0/(\alpha e)) = w$, hence $2x_0 \log(2x_0/(\alpha e)) = L$. Let $f_0(x) = 2x \log(2x/(\alpha e))$. Then $f'_0(x_0) = 2(w + 1)$ and $f''_0(x_0) = 2/x_0$. Lemma 12.1 says

$$\log A_x = f_0(x) + \frac{3}{2} \log x + C_0 + \frac{13/24 + \alpha^2/4}{x} + O(x^{-2}).$$

The displacement $d_0 = O(1)$ cancels the logarithmic and constant residual at x_0 . At $x_0 + d_0$ the remaining term of order $1/x_0$ is

$$\frac{d_0^2 + \frac{3}{2}d_0 + 13/24 + \alpha^2/4}{x_0}.$$

Dividing its negative by $f'_0(x_0)$ gives d_1 . The residual after this correction is $O(x_0^{-2})$. The derivative of $\log A_x$ is asymptotic to $2\log(2x/\alpha)$; this follows either by differentiating the localized integral expansion or by inserting $\log(1+y^2)$ into the saddle calculation. The mean value theorem therefore gives the error in (63). Finally $w = \log x_0 + O(1)$, so (61) tends to $-3/4$. $\square$

Proposition 12.3 (An arbitrary-order inverse procedure). *For each fixed $M \geq 0$, let ξ_M be the sufficiently large solution of*

$$f_0(x) + \frac{3}{2} \log x + C_0 + \sum_{r=1}^M h_r(1)x^{-r} = L.$$

Then

$$x(Y) - \xi_M = O\left(\frac{x_0^{-M-1}}{w+1}\right).$$

The root ξ_M can be expanded to any prescribed algebraic order by Taylor substitution about x_0 , or computed by Newton iteration started at $x_0 + d_0$.

Proof. The truncated logarithm differs from the exact one by $O(x^{-M-1})$ and has derivative asymptotic to $2\log x$, positive for large x . The same holds for the exact logarithm. Existence, uniqueness in a neighborhood of x_0 , and the stated difference follow from monotonicity and the mean value theorem. In that neighborhood the second derivative is $O(1/x_0)$ and the first derivative is bounded below by a positive multiple of $w+1$. Taylor substitution is triangular in successive powers of $1/x_0$; Newton's quadratic error bound supplies the alternative constructive procedure. $\square$

12.2 An exponentially small inverse displacement

An exponential correction must again be measured relative to an exact sector, not relative to a finitely truncated algebraic approximation. Define

$$A_x^{[1]} = 2 \int_0^\infty (1+y^2)^x y e^{-\alpha y} dy, \quad (64)$$

and let $x_1(Y)$ be its real inverse for sufficiently large Y .

Proposition 12.4. *As $Y \rightarrow \infty$,*

$$x(Y) - x_1(Y) = -\frac{3^{-2x_1-2}}{2\log(2x_1/\alpha)} \left(1 + O(x_1^{-1})\right). \quad (65)$$

In particular, the displacement is negative.

Proof. For each fixed odd q , the one-exponential integral has the leading factor $2\Gamma(2x+2)(\alpha q)^{-2x-2}$ and correction $1 + O(1/x)$, by the proof of Lemma 12.1. The sum of the terms $q \geq 5$ is $O(5^{-2x-2})$ relative to the first integral. To bound it directly, use

$$\sum_{q \text{ odd} \geq 5} e^{-\alpha q y} = \frac{e^{-5\alpha y}}{1 - e^{-2\alpha y}} \leq e^{-5\alpha y} \left(1 + \frac{1}{2\alpha y}\right).$$

Both resulting integrals have the same exponential saddle factor, and one has an additional relative factor $O(1/x)$. Consequently

$$\log A_x - \log A_x^{[1]} = 3^{-2x-2}(1 + O(1/x)).$$

The derivative of $\log A_x^{[1]}$ is $2 \log(2x/\alpha) + O(1/x)$. Evaluating at x_1 , and applying Taylor's formula to the inverse, gives (65); the change in the small correction over this displacement is of strictly smaller order. Positivity of all omitted sectors also gives $A_x > A_x^{[1]}$ and hence $x(Y) < x_1(Y)$. $\square$

For integer enumeration the exact counting inverse is $\max\{n : a_n \leq Y\} = \lfloor x(Y) \rfloor$. It is not valid to floor an uncertified asymptotic approximation near an integer. A numerical implementation needing an exact index must enclose $x(Y)$ or compare neighboring exact terms.

13 Reproducible exact checks and numerical diagnostics

The accompanying program `code/verify.py` uses only integer arithmetic, SymPy rational polynomial arithmetic, and mpmath numerical functions. It checks the tangent S-fraction and the conjectured fraction through x^{48} , the Poupard recurrence through row 24, the congruence through $n = 256$, and both determinant products for orders 1 through 6 at $c = 0, 1, 3$. It also derives the correction polynomials through order eight, checks the first saddle coefficient by symbolic differentiation, and compares the moment integral with independent exact binomial sums. These tests support the derivation but do not replace its all-index proofs.

All floating-point tables below use 90 decimal digits internally. They are reproducible diagnostics, not interval enclosures. Exact integer values are used for the fixed-shift and inverse tables. The large- n transition values use positive log-sum evaluation of (43) and (15); they do not use numerical quadrature at a narrow large- n saddle.

13.1 Fixed-shift accuracy

Let $\hat{a}_n^{(r)}$ be the exponential of (24) truncated after $h_r(1)n^{-r}$, and let $r = 0$ mean the known leading equivalent. Table 1 gives $|\hat{a}_n^{(r)}/a_n - 1|$. The table measures a logarithmic-expansion approximation; it is not a comparison of exact odd-pole sectors.

Table 1: Relative errors for fixed-shift logarithmic truncations.

n	$r = 0$	$r = 1$	$r = 3$	$r = 6$
10	0.1059	3.972×10^{-3}	1.575×10^{-5}	5.995×10^{-8}
25	4.464×10^{-2}	6.695×10^{-4}	4.504×10^{-7}	8.724×10^{-11}
50	2.274×10^{-2}	1.703×10^{-4}	2.911×10^{-8}	6.603×10^{-13}
100	1.148×10^{-2}	4.296×10^{-5}	1.849×10^{-9}	5.083×10^{-15}
200	5.765×10^{-3}	1.079×10^{-5}	1.165×10^{-10}	3.943×10^{-17}

13.2 The transition is not already at its limit at moderate size

The explicit window (40) converges relatively slowly because its error contains $(\log n)^2/n$. Table 2 displays that fact rather than hiding it. Its final column is the relative error of

$$1 + \frac{2}{\tau n} + K(\tau)n^{3/2}e^{n\Delta(\tau)} \left(1 + \frac{L_1(\tau)}{n}\right)$$

against the exact positive sum, evaluated at $\tau = \tau_n(w)$. This is Theorem 8.1 truncated with $M = 2$.

Table 2: Explicit-window convergence and a two-contribution correction.

n	w	Normalized moment	$1 + e^{-w}$	Corrected relative error
128	-2	11.11375927	8.389056099	7.178×10^{-4}
128	0	2.715465649	2.0	2.655×10^{-3}
128	2	1.345387446	1.135335283	5.084×10^{-3}
512	-2	9.450352814	8.389056099	4.862×10^{-5}
512	0	2.232398706	2.0	1.971×10^{-4}
512	2	1.192794337	1.135335283	3.634×10^{-4}
2048	-2	8.775283365	8.389056099	3.244×10^{-6}
2048	0	2.076432709	2.0	1.331×10^{-5}
2048	2	1.152262732	1.135335283	2.384×10^{-5}
4096	-2	8.617644401	8.389056099	8.254×10^{-7}
4096	0	2.043674517	2.0	3.389×10^{-6}
4096	2	1.144608013	1.135335283	6.022×10^{-6}

At $w = 0$ the normalized moment is about 2.7155 for $n = 128$, not already two. The uniform corrected formula nevertheless has relative error about 2.66×10^{-3} . At $n = 4096$ that error is about 3.39×10^{-6} . An exact-balance centering such as (42) is preferable when the goal is a precise finite-size coexistence parameter.

13.3 Inverse accuracy and the exact-sector checks

For Table 3 the input is $Y = a_n$, so the true interpolated inverse is exactly n . The reported errors are absolute errors in the index, not relative errors in Y .

Table 3: Absolute inverse errors at the exactly known inputs $Y = a_n$.

n	x_0	$x_0 + d_0$	$x_0 + d_0 + d_1$
10	1.024	1.209×10^{-2}	6.769×10^{-4}
25	0.9517	3.569×10^{-3}	6.945×10^{-5}
50	0.9176	1.481×10^{-3}	1.326×10^{-5}
100	0.8934	6.317×10^{-4}	2.648×10^{-6}
200	0.8753	2.751×10^{-4}	5.471×10^{-7}

The verification output additionally records, at $n = 8, 16, 32$, the exact integer moment divided by F_n , the exact first polynomial sector, the $q = 3$ sector, and the positive residual bounded by (20) with $Q = 5$. This is a useful way to test the beyond-all-orders normalization without subtracting two low-order algebraic approximations. The symbolic and numerical results, precision, package versions, and all displayed table data are in `data/verification.json`.

13.4 Complex roots and additional universality checks

Table 4 follows the root associated with $w = \pi i$ in Theorem 11.1, using the affine coordinate from (40). The coefficients supplied to reverse Horner evaluation are exact integers. Root solving is numerical; the normalized residuals are below 10^{-55} at the working precision, but that is not an

interval certificate of root location or uniqueness. The all-large- n existence and simplicity assertion comes from the theorem, not from these residuals. The limiting real and imaginary parts are 0 and π , respectively.

Table 4: The zero associated with $w = \pi i$ in the coexistence coordinate.

n	$\Re w_{n,0}$	$\Im w_{n,0}$
32	0.6034631023	4.791335302
64	0.593852504	3.93115992
128	0.4184735045	3.549529939
256	0.2728574154	3.35895883

The JSON output also tests the general transition on gamma moment models with $(p, b) = (3/2, 1/4), (2, 1), (3, 2)$, where b is the gamma shape parameter. These include negative and positive logarithmic displacements and an endpoint error of order $n^{-1/2}$ when $p = 3/2$. At $n = 4096$ and $w = 0$, their normalized moments are approximately 2.00822, 2.00774, and 2.03211, respectively, approaching the common limit two. These diagnostics supplement, rather than establish, Theorem 10.1.

14 Further research questions and a formalization route

The following questions are proposed research directions, not conclusions of the present proofs. They are chosen to extend a structural result or remove a specific limitation, rather than merely ask for more terms of a calculation whose general coefficient algorithm is already given.

Research question 1 (A direct combinatorial explanation of the new fraction). Can one give a weight-preserving bijection from the appropriate Poupard objects, or their marked variants, to Dyck paths with successive weights 9, 8, 25, 24, ...? Theorem 3.3 proves equality through formal contraction. A bijection would explain the squared-odd and adjacent-even weights at the level of objects rather than generating functions.

Research question 2 (Which shifts admit polynomial S-weights?). For which real or integer shifts c does the J-fraction (11), or a fixed Christoffel transform of its moment measure, admit S-weights that are polynomial functions of the index on each parity class? The tangent and Poupard examples suggest a classification problem. A complete answer should distinguish positivity of weights, existence of an algebraic contraction, and a combinatorial interpretation.

Research question 3 (Minimal modular descriptions). Corollary 4.2 gives an elementary finite-height bound. What is the minimal rational denominator modulo m , and what are the minimal preperiod and period of $A_n(c)$? The modulus-nine collapse is much smaller than the generic height bound. Prime-power lifting and cancellation of path products may produce comparably sharp formulas for other moduli, but those formulas have not been proved here.

Research question 4 (A local limit theorem across coexistence). Can the two-point weak limit and conditional Gaussian law be upgraded to a uniform lattice local limit theorem, including the exact probability of each fixed small index and Edgeworth corrections in the macroscopic window? Such a result must match an endpoint series to a lattice Gaussian rather than approximate both components by one continuous law. The explicit E_m and L_m are natural inputs but do not by themselves establish a local theorem.

Research question 5 (Growing-index zero arrays and higher corrections). Theorem 11.1 determines zeros corresponding to each fixed $w_j = (2j + 1)\pi i$. How large may $j = j(n)$ grow while a uniform analogue remains valid, and what are the next corrections to their locations? Determine whether this local array joins a global limiting zero curve. The positivity/normal-family proof gives qualitative convergence on fixed compact sets, not quantitative bounds on expanding complex domains.

Research question 6 (The boundary $p \downarrow 1$). In Theorem 10.1, the critical stationary point approaches the endpoint as p tends to one. Determine a joint $n \rightarrow \infty$, $p \downarrow 1$ scaling that connects the separated first-order transition to its endpoint-coalescence limit. Neither the Gaussian expansion nor the fixed- p error estimate is uniform in this joint limit.

Research question 7 (Slowly varying and competing tails). Extend (47) to a slowly varying multiplier or to several competing exponential scales. A multiplier depending on n at the saddle should alter the balance by its logarithm, but a theorem should specify sufficient regularity, uniform remainder bounds, and the possibility of more than two competing components. The exact odd-pole decomposition offers a concrete positive model for testing such a theory.

Research question 8 (Certified computation and inverse rounding). Develop explicit constants and computable thresholds in Theorem 8.1, then combine them with interval arithmetic to certify the coexistence parameter and integer inverse. The sector bound (20) is already explicit, but the all-order Laplace remainders in this article are asymptotic bounds with unspecified constants. A certified implementation must close that gap rather than interpret high-precision diagnostics as enclosures.

14.1 A realistic Lean route

The finite algebraic layer can be separated from the analytic layer. A first formalization target is the path recurrence for T_n , formal S- and J-fraction truncations, and their coefficient stabilization. The contraction lemma and binomial translation then give Theorem 3.3. Weighted Motzkin paths modulo nine give Theorem 4.1 with a small, local divisibility argument. These steps need no asymptotic integration.

A second layer can formalize the gamma cancellation identity, the finite polynomial H_n , and the elementary odd-tail integral bound. The positive integral representation is the bridge to that layer. The positive-coefficient complex bound and a normal-family theorem would then support the complex zero result. The most substantial separate obligations are a parameter-uniform Laplace theorem with an endpoint component, differentiation and inversion of its asymptotics, and the conditional central limit argument. The present Python checks do not discharge those obligations. A future formalization should record precisely which theorems are kernel checked, which numerical values are interval certified, and which results remain ordinary mathematical proofs.

15 Conclusion

The central Poupard sequence provides two concrete, now-proved answers to statements displayed as conjectural in OEIS: the December 2025 continued fraction and the uniform 3-valuation observation. Their proofs are short because the right formal representation exposes the relevant structure.

The larger contribution of the investigation is a coherent analytic picture. Fixed shifts possess exact positive exponential sectors with explicit tail control. Quadratic shifts create an endpoint–saddle competition, a logarithmically displaced coexistence window, and a macroscopic two-component

index law. Positivity promotes the real limit to an analytic limit and rigorously locates nearby complex zeros. The same mechanism persists for a large class of exponentially decaying moment measures, while the power in their tail determines the logarithmic displacement. Lambert inversion then connects the fixed-shift transseries to a usable index estimate, without confusing algebraic truncation errors with exponentially small sectors.

The known leading equivalent is preserved and attributed; the extensions are accompanied by proofs, coefficient algorithms, tests, and clearly stated limitations. The next substantial gains would come from growing-index complex zero arrays, joint endpoint limits, local probability laws, or fully certified remainder estimates, rather than from finite-data extrapolation.

A Coefficient generation and reproduction

The formal logarithm coefficients used in the code may be generated without symbolic logarithm expansion. If $C_0 = 1$ and $\sum C_r t^r = \exp(\sum_{r \geq 1} \ell_r t^r)$, differentiating gives

$$\ell_r = C_r - \frac{1}{r} \sum_{j=1}^{r-1} j \ell_j C_{r-j}.$$

Together with (22) and (25), this is an exact rational-polynomial algorithm at every finite order. The delivered verification file contains C_r , ℓ_r , and the logarithmic h_r through $r = 8$; in that file the common symbolic argument is $z = c\alpha^2$. The notation $h_r(c)$ in the article already includes that substitution.

To reproduce the supplied full diagnostics from the package directory:

```
python -m pip install -r requirements.txt
python code/verify.py --max-n 4096 --dps 90
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

A shorter diagnostic run can be directed to a separate directory:

```
python code/verify.py --max-n 512 --output-dir data/quick
```

The PDF can be rebuilt without rerunning Python; its generated table fragments are included. No external font files are bundled. The exact tests use independent recurrences, contractions, determinant calculations, and symbolic derivative identities where practical. The scripts contain no theorem-prover certificates and do not label floating-point output as a rigorous error enclosure.

[Write note, batch 7301, 1 October 2026.] In the collection the make file is `code/Makefile` and the requirements file is `data/requirements.txt`. Its targets assume the report root as the working directory, so it is invoked as `make -f code/Makefile`. The full command above rewrites the six generated files in the shipped `data/`, because `code/verify.py` writes by default to the `data/` directory beside `code/`; the `verify` target does the same and also overwrites `data/verification_run.txt` with its standard output. Run these commands on a copy of the report directory, or pass `-output-dir` with a scratch directory. At intake the full command was rerun on a copy: all six generated files were equal to the shipped ones apart from Windows line ends, and the standard output differed only in the line naming the output directory. That line in the shipped `data/verification_run.txt` names the authors' working directory `/mnt/data/...`, which is not part of the collection.

B Source and novelty audit

The source review checked the OEIS entries available at inspection for A125054, A125053, A000182, and A000364; Bala’s continued-fraction note; the primary Foata–Han finite-difference paper; the indicated NIST DLMF sections; and the ProveIt repository material described in Section 1. Targeted searches for the central Poupard asymptotics, shifted tangent moments, and transition terminology did not identify a matching publication for the full package of extensions. Such a search is not proof of publication novelty. In particular, the proofs use standard Laplace and continued-fraction methods, and specialized consequences may have precedents under other terminology.

Repository code searches for A125054 and A000182 returned no matches in the inspected search results. Search indexing and unexamined material prevent this from being an exhaustive nonduplication certificate. The earlier snapshot was kept fixed even though the repository’s main branch advanced during the investigation. Full details and source URLs are recorded in `PROVENANCE.md`.

The verification boundary is exact: the continued-fraction and congruence statements have all-index mathematical proofs; the analytic theorems have ordinary mathematical proofs with stated asymptotic errors; finite exact computations and floating diagnostics are supplementary. No claim of independent peer review, Lean verification, or interval certification is made.

C Provenance of the collection edition

[Write note, batch 7301, 1 October 2026.] This appendix was added in the write.

Source. The report is built from one manuscript, number 47 of batch 7301 of ProveIt’s incoming-reports intake: the archive `Central_Poupard_OEIS.zip` (16 files, a 28-page PDF), which arrived in commit `3a9518c52` and was placed in the collection by commit `9df4ba51a`. Its pinned repository commit is `ef66df0fdb8fc64cbc80756d622cab0e8e85e262` (1 October 2026); `PROVENANCE.md` also records a later read of `dc1a7242d2ab4a4496b7dfee63256ee767b21fc9`, not used for content. The author line is kept as delivered and names no assistant; like every manuscript of the intake, the report is AI-assisted and unrefereed.

What the write changed. Every one of the 108 delivered labels received the prefix `pou:`, and every reference was updated with it; this appendix adds one label. The write added the `[Write note]` paragraphs (after the abstract, in Sections 1 and 7, in Appendix A and here), the bibliography entries [15] and [14], and the `\writenote` macro. The `\input` and `\tableinput` paths `data/...` are unchanged. No delivered sentence, statement, proof or table was changed or removed.

Delivered names and shipped paths. `article.tex`, `README.md` (rewritten as the collection’s report guide) and `PROVENANCE.md` are at the report root; `code/verify.py` and the seven files of `data/` kept their paths; the `Makefile` moved from the package root to `code/`, and `requirements.txt` and `BUILD_REPORT.json` from the package root to `data/`. All but the article and the `README` are byte-identical to the delivery. Not shipped: the delivered `article.pdf` (replaced by a build of this edition) and the checksum ledger `SHA256SUMS` (verified 15/15 at intake, then retired). `data/BUILD_REPORT.json` describes the delivered 28-page PDF and the delivered source, which are no longer the shipped files.

Relation to repository material. No other repository report or formal development treats A125054 or its analytic model. The Lambert-chart transseries package [14] is a methodological precedent only, and the shared constant s_* with the A277364 report [15] is discussed in the notes in Sections 1 and 7. None of the statements of this report is formalized.

References

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- [14] [Added in the write.] The package of [13] in the repository: `Analysis/Transseries/docs/series-and-transseries/Logarithmic_Critical_Endpoint_Lambert_Charts/`.
- [15] [Added in the write.] ProveIt research-report collection, *Asymptotics of OEIS A277364*, `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a277364-bell-asymptotics/A277364_asymptotics.tex`, equation `eq:taulambert`.